The two-dimensional amplitude integral and its one-variable density

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Abstract

For an amplitude $\Phi(u,v,s)$ the two-dimensional chart integral $Z_\Phi(N) = \int_0^b u^{h_1} \int_0^b v^{h_2} e^{-\beta(Nu^{k_1}v^{k_2})^2} \Phi(u,v,Nu^{k_1}v^{k_2})\,dv\,du$ is symmetric under exchange of the coordinates, and when $\Phi(u,v,s) = \Psi(u,s)$ depends on u only, with equal exponents $(h_1+1)/k_1 = (h_2+1)/k_2 = p$, it is the transfer integral of the exact one-variable density $\rho_\Psi(r,s) = k_2^{-1} r^{p-1} \int_{(r/b^{k_2})^{1/k_1}}^b u^{-1}\Psi(u,s)\,du$ over $(0,R]$, $R = b^{k_1+k_2}$. This is the mechanism behind the anchored subtraction of the $d=2$ theorem (Astra's step 4). Zero `sorry`/`axiom`.

1 The things formalised

```
noncomputable def twoDAmp (  b N :  ) (h  h  k  k  :  ) (Φ :   →   →   → ) :   :=
    u in Ioc (0 :  ) b, u ^ h  *   v in Ioc (0 :  ) b, v ^ h
    * (Real.exp (-  * (N * (u ^ k  * v ^ k )) ^ 2) * Φ u v (N * (u ^ k  * v ^ k )))
noncomputable def uDensity (p b :  ) (k  k  :  ) (Ψ :   →   → ) (r s :  ) :   :=
  1 / (k  :  ) * r ^ (p - 1)
    *   u in Icc ((r / b ^ k ) ^ ((k  :  ) ¹)) b, u ¹ * Ψ u s
theorem twoDAmp_swap ... :
    twoDAmp   b N h  h  k  k  Φ = twoDAmp   b N h  h  k  k  (fun v u s => Φ u v s)
theorem twoDAmp_uOnly_eq_transferZ (  b N p :  ) (h  h  k  k  :  ) (h  : 0   ) (hb : 0 < b)
    (hN : 0   N) (hk  : 0 < k ) (hk  : 0 < k ) (hp  : ((h  :  ) + 1) / k  = p)
    (hp  : ((h  :  ) + 1) / k  = p) (Ψ :   →   → )
    (hΨc : Continuous (Function.uncurry Ψ)) :
    twoDAmp   b N h  h  k  k  (fun u _ s => Ψ u s)
      = transferZ (uDensity p b k  k  Ψ)   (b ^ (k  + k )) N
```

2 Proof strategy

Inner substitution $r = u^{k_1}v^{k_2}$ (units 44's two substitution lemmas) turns the v-integral into $k_2^{-1}(u^{k_1})^{-p}\int_0^{u^{k_1}b^{k_2}} r^{p-1}e^{-\beta(Nr)^2}\Psi(u,Nr)\,dr$, and $u^{h_1}(u^{k_1})^{-p} = u^{-1}$ by the equal-exponent relation. The resulting iterated integral over the curved region $\{r \le u^{k_1}b^{k_2}\}$ is written with an indicator on the rectangle $(0,b] \times (0,R]$ and swapped by `integral_integral_swap`; the u-section is the integral over $[(r/b^{k_2})^{1/k_1}, b]$ because $r \le u^{k_1}b^{k_2} \iff (r/b^{k_2})^{1/k_1} \le u$ (`Real.rpow_le_rpow_iff`). Integrability on the rectangle follows from `integrable_prod_iff`: the r-sections are dominated by $u^{-1}Mr^{p-1}$ on $(0, u^{k_1}b^{k_2}]$, whose integral is $Mb^{k_2p}u^{pk_1-1}/p$, integrable in u since $p > 0$.

3 Roadblocks and resolutions

- `gcongr` side goals $0 \le u^{-1}$, $0 \le r^{p-1}$ need $0 < u$, $0 < r$ as named hypotheses in context (membership in `Ioc` is not enough for `positivity`).

- `integrableOn_rpow_sub_one_Ioc` (unit 49) is only on $(0,1]$; the general interval comes from `intervalIntegral.intervalIntegrable_rpow'` via its `.1` projection.
- `positivity` cannot prove $0 \le Nr$ from $0 \le N$ and $r \in \mathrm{Ioc}$; use `mul_nonneg` explicitly.

4 Outcome

The exact density of a one-variable amplitude is available in transfer form. Next: expand ρ_Ψ (Piece A: $\Psi(0,s)\,k_1^{-1}\log(R/r)$ plus the axis integral $I_\Psi(s) = \int_0^b (\Psi(u,s) - \Psi(0,s))/u\,du$ with an $O(r^{1/k_1})$ tail), bound the mixed term uvG by a shifted constant-kernel box integral, and assemble the $d = 2$ theorem $Z(n) = n^{-p/2}(\frac{A}{2}\log n + B) + O(n^{-(p+\delta)/2}(1 + \log n))$.